[Checkpoint 5] Running full pipeline on sample query...
  Q: What are the immediate steps an agent should take when a flight is delayed due to a mechanical issue?
  A: When a flight is delayed due to a mechanical issue, the gate agent must take the following immediate steps:

1. Immediately notify the Maintenance Control Center (MCC) via the Operations Portal using the AOG (Aircraft on Ground) notification form. Include the tail number, departure gate, scheduled departure time, and symptom description provided by the flight crew.

2. Enter delay code 41 (Aircraft Defect) or 43 (Aircraft Scheduled Maintenance) in SABRE provisionally. The MCC will confirm or update the code once diagnosis is complete.

3. Announce to passengers at the gate: "We are currently experiencing a maintenance review on your aircraft. We expect an update within 15 minutes. We apologize for the inconvenience."

4. Do NOT speculate on the nature of the mechanical issue or estimated repair time to passengers until the MCC provides an official estimate.

  Sources: airline_kb/Delay_Classification.txt

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RUNNING ALL TEST QUERIES
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Q: What are the immediate steps an agent should take when a flight is delayed due to a mechanical issue?
A: When a flight is delayed due to a mechanical issue, the gate agent must take the following immediate steps:

1. **Notify Maintenance Control Center (MCC)**: The agent must immediately notify the MCC via the Operations Portal using the AOG (Aircraft on Ground) notification form. This notification should include the tailnumber, departure gate, scheduled departure time, and a symptom description provided by the flight crew.

2. **Enter Delay Code**: The agent must enter delay code 41 (Aircraft Defect) or 43 (Aircraft Scheduled Maintenance) in SABRE provisionally. The MCC will confirm or update the code once the diagnosis is complete.

3. **Announce to Passengers**: The agent should announce to passengers at the gate: "We are currently experiencing a maintenance review on your aircraft. We expect an update within 15 minutes. We apologize for the inconvenience."

4. **Do Not Speculate**: The agent must not speculate on the nature of the mechanical issue or estimated repair time to passengers until the MCC provides an official estimate.

These steps ensure proper communication and documentation of the delay while maintaining compliance with operational procedures.

Sources: airline_kb/Delay_Classification.txt

Q: Under what weather conditions does the airline automatically rebook passengers on the next available flight?
A: Passengers are automatically rebooked on the next available flight when a Weather Advisory is issued and any of the following conditions apply:

1. Flight delay is forecast to exceed 2 hours due to weather at the departure airport.
2. Flight delay is forecast to exceed 3 hours due to weather at the destination airport.
3. Flight is in the cancellation queue based on OCC recovery planning.

Automatic rebooking does not trigger for forecasted weather that has not yet resulted in an issued Weather Advisory.

Sources: airline_kb/Weather_Operations.txt

Q: What is the maximum crew duty period allowed before a mandatory rest period is required?
A: The maximum crew duty period allowed before a mandatory rest period is required varies based on the time of day at the start of duty. For a 2-pilot crew with a single flight segment, the maximum Flight Duty Period (FDP) is as follows:

- 0000–0459: 9 hours
- 0500–0659: 13 hours
- 0700–1359: 14 hours
- 1400–2059: 13 hours
- 2100–2359: 12 hours

After any flight duty period, a crewmember must receive a minimum rest period of 10 hours, which includes 8 consecutive hours of sleep opportunity.

Sources: airline_kb/Crew_Rest_Regulations.txt

Q: Explain the priority order for gate reassignment during irregular operations.
A: During irregular operations (IROPS), the priority order for gate reassignment is as follows:

1. **Priority 1**: Inbound flight with a medical emergency or diverted flight.
2. **Priority 2**: Flights with connecting passengers whose connection time is within 45 minutes of the minimum connection time (MCT).
3. **Priority 3**: Outbound flights with the highest number of confirmed revenue passengers.
4. **Priority 4**: Flights where crew flight time regulations will be violated if departure is delayed more than 60 minutes.
5. **Priority 5**: Cargo-only operations or empty aircraft repositioning.

When two flights have equal priority ranking, the flight with the earlier scheduled departure time takes precedence. The Gate Control Supervisor in the Operations Control Center must approve all Priority 1 and 2reassignments, while Priority 3–5 reassignments may be approved by the Hub Station Manager.

Sources:

Q: How should ground staff communicate a delay of more than 2 hours to connecting passengers?
A: Ground staff should communicate a delay of more than 2 hours to connecting passengers by providing proactive passenger communication and coordinating with Customer Service for early rebooking options. This includes notifying passengers at the gate within the required timeframes and ensuring that automated notifications are sent via the app or SMS as appropriate.

Sources: airline_kb/Delay_Classification.txt

Q: What documentation is required when a flight is cancelled due to Air Traffic Control (ATC) restrictions?
A: When a flight is cancelled due to Air Traffic Control (ATC) restrictions (delay code 81–89), the following documentation is required:

- FAA/ATC notification printout or ACARS message (proving ATC cause)
- OCC IROPS log entry with timestamp of ATC notification
- SABRE cancellation record with delay code 83 (ATC Flow Control) or 81 (ATC Staffing)
- Passenger rebooking log with waiver codes applied

Sources: airline_kb/Delay_Classification.txt

Q: Compare the rebooking options available to elite status passengers versus standard passengers during a weather cancellation.
A: During a weather cancellation, elite status passengers have significantly more favorable rebooking options compared to standard passengers.

Rebooking Options for Elite Status Passengers:
1. **Automatic Rebooking:** Elite passengers (Platinum, Gold, and Silver) are prioritized in the automatic rebooking process, with Platinum Elite passengers receiving the highest priority.
2. **Rebooking Window:**
   - Platinum Elite: 7 days from original departure.
   - Gold Elite: 72 hours from original departure.
   - Silver Elite: 48 hours from original departure.
3. **Available Flights:** Elite passengers can be rebooked on any available seat, including partner airlines (Gold and Platinum) or any carrier (Platinum) if the airline cannot provide service within 3 hours.
4. **Hotel Accommodation:** Gold and Platinum Elite passengers are entitled to hotel accommodations for any overnight delay, regardless of the cause, while Silver Elite passengers receive accommodations only forcarrier-caused delays.
5. **Meal Vouchers:** Elite passengers receive higher meal vouchers based on the duration of the delay (e.g., Platinum Elite receives $60 for delays of 1-2 hours, while Standard passengers receive only $15).
6. **Change Fee Waiver:** Elite passengers enjoy unlimited changes without fees for a specified period (90 days for Platinum, 60 days for Gold), while Standard passengers have a fee waiver only for the rebooked flight.

Rebooking Options for Standard Passengers:
1. **Automatic Rebooking:** Standard passengers are also automatically rebooked but are placed at the bottom of the priority queue.
2. **Rebooking Window:** 24 hours from original departure.
3. **Available Flights:** Standard passengers can only be rebooked on any available seat in the same cabin class.
4. **Hotel Accommodation:** Not provided unless the overnight delay is caused by the carrier.
5. **Meal Vouchers:** Standard passengers receive lower meal vouchers (e.g., $15 for delays of 1-2 hours).
6. **Change Fee Waiver:** Standard passengers have a change fee waiver only for the rebooked flight.

In summary, elite status passengers benefit from a broader range of rebooking options, higher priority, and additional amenities compared to standard passengers during weather cancellations.

Sources:

Q: What are the de-icing requirements for departures in freezing rain conditions?
A: In freezing rain conditions, the de-icing requirements for departures are as follows:

1. Aircraft must receive a two-step treatment: Type I fluid application first to remove existing ice, followed immediately by Type IV fluid application for anti-icing protection.
2. De-icing must be performed as close to departure time as possible. The holdover time clock begins when the Type IV application is complete.
3. The holdover time for Type IV fluid in freezing rain conditions at 28°F is approximately 14–26 minutes. Departure must occur within this holdover window.
4. If the aircraft does not depart within the holdover window, a full re-application of Type I followed by Type IV is required before takeoff.
5. The Captain must perform a pre-takeoff contamination check (visual inspection of wings from cabin window or via wing camera if equipped) before initiating the takeoff roll.
6. Gate agents must not close the aircraft door until the Captain confirms that de-icing is complete and the anti-icing fluid has been applied.

Sources: airline_kb/Weather_Operations.txt

Q: Describe the escalation process when a mechanical delay extends beyond the originally estimated time.
A: When a mechanical delay extends beyond the originally estimated time provided by the Maintenance Control Center (MCC), the following escalation process must be followed:

1. The gate agent must immediately notify the OCC Duty Manager that the estimated completion time (ECT) has been missed.
2. The Duty Manager will then contact the MCC for a revised ECT and determine if an aircraft swap is now required.
3. If the revised ECT adds more than 60 minutes beyond the original estimate, the Duty Manager must notify the Customer Service Director.
4. Gate agents are required to update the passenger announcement every 15 minutes (instead of the standard 30 minutes) once an ECT has been missed.
5. If passengers have boarded and waited more than 3 hours on the aircraft without takeoff, they must be offered the option to deplane if the aircraft has boarded. Additionally, these passengers are subject to the DOT Tarmac Delay Rule, and the Captain must be informed immediately while the OCC initiates the Tarmac Delay Contingency Plan.

Sources: airline_kb/Delay_Classification.txt

Q: How does crew rest regulation compliance affect flight scheduling recovery during a major hub disruption?
A: Crew rest regulation compliance significantly affects flight scheduling recovery during a major hub disruption (IROPS Level 3). When crew rest limits are reached or violated, it becomes a primary constraint on recovery planning. Specifically, if inbound crew members have been delayed (for example, by weather) for more than four hours, their Flight Duty Period (FDP) clock may be close to its maximum, preventing them from being assigned additional flying, even if the aircraft is ready. This necessitates the activation of reserve crew, and if local reserves are exhausted, the OCC Crew Desk must contact regional crew bases for reassignment, which may involve positioning flights to bring crew to the disrupted hub.

Moreover, proactive management of crew rest is essential; recovery planning for crew begins as soon as a disruption is detected to minimize the schedule impact. This includes checking the remaining FDP availability, rest period status, cumulative flight time tracking, and reserve crew availability. Thus, adherence to crew rest regulations is crucial for effective recovery during major disruptions.

Sources: airline_kb/Crew_Rest_Regulations.txt, airline_kb/IROPS_Procedures.txt
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